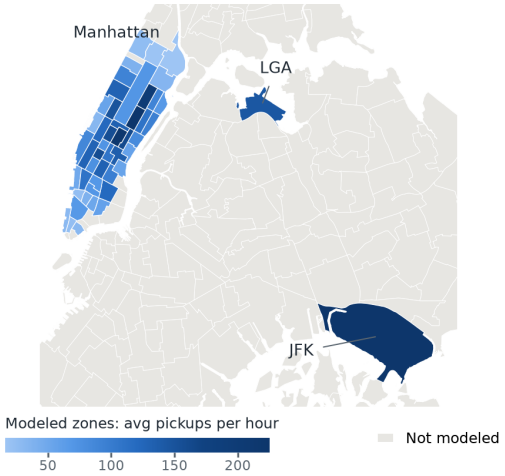
NYC Yellow-Taxi Demand: Model Table at a Glance

`data/model_table.csv.gz` · built by `build_dataset.py` · snapshot of 2 Nov 2026 · reproduce everything with `python build_dataset.py`

Task. Predict how many yellow-taxi pickups happen in each busy NYC taxi zone during each hour, **24 hours in advance**. **One row = one zone × one hour.** The target column is **pickups** (trips starting in that zone in that hour).

Rows	1,504,741
Period	29 Jan 2023 – 31 Aug 2026 (hourly, NYC local time)
Zones	49: 47 in Manhattan + JFK Airport + LaGuardia Airport (map)
Pickups / row	mean 93, median 62, max 1,167; 1.2% of rows are 0
dev split	29 Jan 2023 – 31 Aug 2025, 1,075,501 rows: model selection with time-ordered CV
test split	1 Sep 2025 – 31 Aug 2026, 429,240 rows: used once, at the end (a full year)



Columns

Group	Columns	What it means	Source
IDs (not features)	`zone_id, zone_name, borough, ts_hour, split`	Row identifiers; *split* = dev / test	—
Location	`zone_code, borough_code`	Zone and borough as integer categories	TLC zone lookup
Recent demand	`lag_h, lag_48h, roll_24h, roll_7d`	Pickups 24h and 48h earlier; average over the 24h / 7 days ending 24h earlier	TLC trips
Weekly pattern	`lag_168h, lag_336h, same_hour_4wk`	Same hour 1 and 2 weeks earlier; average of the same hour over the last 4 weeks	TLC trips
Trip context	`duration_lag_h, fare_lag_h`	Average trip duration (min) and fare ($) in the zone 24h earlier; empty if there were no trips	TLC trips
Calendar	`hr, dow, month, is_weekend`	Hour 0–23, day of week (0 = Sunday), month, weekend flag	Timestamp
Weather	`temp_c, feels_like_c, precip_mm, snow_depth_m, wind_kmh, cloud_cover_pct, is_precip, likely_snow`	Midtown weather for the target hour; is_precip = precip > 0.1 mm; likely_snow = precip at 1 °C or colder	Open-Meteo (gfs_global)
Holidays	`is_holiday, is_holiday_season, days_to_holiday, days_since_holiday`	Offices closed (NYSE calendar); Dec 24 – Jan 1 and the day after Thanksgiving; days to the next / since the last holiday (capped at 7)	holidays pkg
Weather alerts	`alert_winter, alert_heat, alert_cold, alert_wind, alert_flood`	Highest NWS alert in effect for the hour, counted only if it was already issued 24h earlier: 0 none, 1 advisory, 2 watch, 3 warning. In Manhattan, winter-warning hours ran at 42% of normal demand	NWS via IEM
Test in CV	`post_cbd_pricing, is_minor_holiday`	Congestion-pricing period (from 5 Jan 2025); holidays most offices ignore (Columbus, Veterans, Election Day). Possibly redundant, so compare models with and without them	—

blue = planned model features amber = to be tested in CV white = identifiers. **Not in the file:** `hist_avg` (average pickups per zone × weekday × hour). Compute it inside each CV fold from that fold's training rows only, otherwise it leaks the answer.

Where the data comes from

Source	What we take	Access
NYC TLC trip records	Every yellow-taxi trip, Jan 2023 – Aug 2026 (157M valid trips), aggregated to pickups per zone per hour. Zones with no trips in an hour get 0. Zones kept only if they averaged at least 20 pickups/h from 5 Jan 2025 to 31 Aug 2025 (dev data only).	Free monthly Parquet files from the TLC website
Open-Meteo Historical Forecast API	Hourly weather at one Midtown point from the **gfs_global** weather model. This is an archive of past forecasts, so it is very close to the observed weather. Project 2 uses the live Open-Meteo forecast with the same model and point, so training and deployment come from one source.	Free, no API key; cached in `data/raw/`
NWS weather alerts	Every watch, warning and advisory for Manhattan, Southern Queens (JFK) and Northern Queens (LGA), with the time it was first issued. The Iowa Environmental Mesonet archive provides the history; Project 2 uses the live api.weather.gov feed, which carries the same alert codes.	Free, no API key; cached in `data/raw/`
Python *holidays* package	Stock-exchange (NYSE) holiday calendar as the "offices closed" list, plus other NY public holidays. Known years ahead, so no leakage.	Open source, offline (v0.107)

Good to know before modeling

- **No look-ahead.** Every demand feature is at least 24h old, so each row only uses what was known 24h before.
- **Weather is treated as a perfect forecast.** Rows hold the weather that actually occurred, not the forecast from the day before, so test scores are slightly optimistic (mostly because of rain). Project 2 will measure the gap.
- **Cross-validate in time order.** Use folds ordered by time (e.g. by month), not shuffled K-fold, and build the folds on timestamps so all zones for an hour stay in the same fold.
- **Holiday echo.** Weekly lags carry a holiday's low demand into the following week (e.g. lag_168h on the Thursday after Thanksgiving). Watch for this in the per-day errors.
- **Data gap removed.** TLC's raw data is missing most trips from 2023-09-21 13:00 to 2023-09-25 00:00. Those hours and the 4 weeks after them (whose lags look back into the gap) are dropped, 36,995 dev rows.